

Introduction

There are some useful scripts.

vcf2snpbinner.R

A R script for transforming vcf file to [SNPBinner](#) input file. The heterozygosity and missing rate will be calculated for output SNP marker.

Usage

```
Rscript ./vcf2snpbinner.R -h
```

```
usage: vcf2snpbinner.R [--] [--help] [--opts OPTS] [--input INPUT]
      [--out OUT] [--parent1 PARENT1] [--parent2 PARENT2] [--minDP_p1
      MINDP_P1] [--minDP_p2 MINDP_P2] [--max_missing MAX_MISSING]
```

a program for converting vcf to table of snpbinner. genotype same as parent_1 is designated 'a', genotype same as parent_2 is designated 'b', heterozygous genotype is designated 'h', missing genotype is designated '-'

flags:

-h, --help show this help message and exit

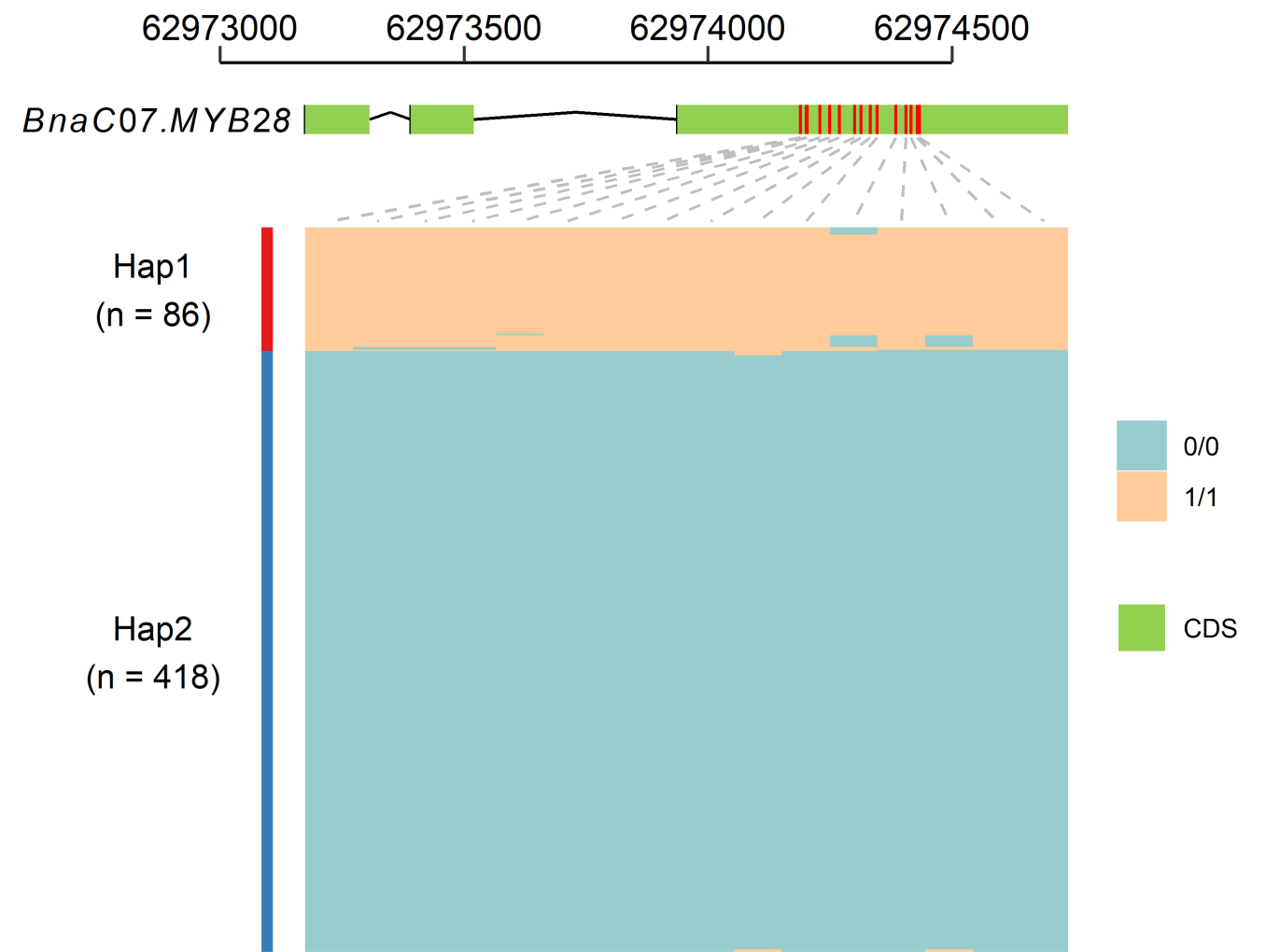
optional arguments:

-x, --opts RDS file containing argument values
-i, --input vcf or vcf.gz file containing two parents and
 progeny lines
-o, --out output file prefix
-p, --parent1 name of parent_1
-P, --parent2 name of parent_2
-d, --minDP_p1 Minimum depth of parent_1 [default: 5]
-D, --minDP_p2 Minimum depth of parent_2 [default: 5]
-m, --max_missing Maximum missing rate of SNP [default: 0.3]

Haplotype.R (GeneStructure with Variation)

A R script for drawing gene structure and the variation of this gene in a population. A gtf file containing target gene, a vcf file containing variation of this gene and phenotype data is needed. Hierarchical clustering algorithm was adopted to distinguish different haplotype, the number of haplotype can be designated

according clustering result. Some polymorphism may exist within samples belonging to the same haplotype, you can divided them into different haplotypes by setting more haplotypes.



ePCR.pl

A perl script for ePCR. Input is tsv (tab-separated values) file containing three columns (PrimerID, forwardPrimer, Reverse Primer).

Requiemment

- [ePCR](#)

Preparation

```
# lower letter to UPPER letter
seqkit seq -u reference.fa > genome.fa
# prepare sequence database for re-PCR searches
famap -t N -b genome.famap genome.fa
fahash -b genome.hash -w 12 -f 3 genome.famap
```

Usage

```
perl ePCR.pl -h
```

```
#####
#
# Usage:  ePCR.pl --input primer.txt --output output.txt
#
# Required:
#
#   --input <string>          input filename, one pair primer per line, tab
seperated, e.g.:
#                               primerID    Left_primer_seq Right_primer_seq
#
#   --output <string>         output filename.
#
#####
```

slidingWindow.R

A sliding window function in R. The R package `tidyverse` should be installed. `values` is a vector containing column names which need be calculated.

```
source("./slidingWindow.R")
# An example
sldwid <- slidingWindow(df = df, winSize = 1000000, winStep = 200000, groups =
"CHROM", position = "POS", values = c("R.R3.depth", "R.qY.depth"), fun = "mean")
```

addUp.R

A R function for calculating accumulation value for a column of a table. For example, a data.frame contain two columns, "chromosome" and "position", this function will calculate the accumulation position of different chromosome, then a list containing a table with a new column "position_add_Up", a vector containing breaks position, a vector containing labels, a vector containing gaps position, will be returned.

```
source("./addUp.R")
# An example
addUp(df = df, len = len, group = "chromosome", pos = "position", band = 0.01)
addUp(df = df, len = len, group = "chromosome", pos = c("start", "end"), band =
0.01)
```

run_DESeq2.R

A R script for differential expression analysis using DESeq2 (with biological replication). You need to prepare three files:

Requiemment

- **read count matrix file.**
- **samples file**, tab-delimited text file indicating biological replicate relationships. e.g.

```
cond_A  cond_A_rep1
cond_A  cond_A_rep2
cond_B  cond_B_rep1
cond_B  cond_B_rep2
```

- **contrasts file**, tab-delimited text file containing the pairs of sample comparisons to perform. e.g.

```
cond_A  cond_B
cond_Y  cond_Z
```

Usage

```
Rscript run_DESeq2.R -h
```

```
usage: run_DESeq2.R [--] [--help] [--opts OPTS] [--matrix MATRIX]
      [--samples_file SAMPLES_FILE] [--min_reps MIN_REPS] [--min_cpm
      MIN_CPM] [--contrasts CONTRASTS]
```

Run differential expression analysis using DESeq2.

flags:

-h, --help show this help message and exit

optional arguments:

-x, --opts RDS file containing argument values
-m, --matrix matrix of raw read counts (not normalized!)
-s, --samples_file tab-delimited text file indicating biological
 replicate relationships.
--min_reps At least min count of replicates must have cpm
 values > min cpm value. [default: 2]
--min_cpm At least min count of replicates must have cpm
 values > min cpm value. [default: 1]

```
-c, --contrasts    file (tab-delimited) containing the pairs of
                   sample comparisons to perform.
```

extractSeq.sh

If there is a gene/transcript id you are interested and the corresponding genomics data, then you want to obtain genomic, CDS, pep sequence and gene structure information of this gene/transcript, you can use this shell script.

Usage

```
#chmod u+x extractSeq.sh
./extractSeq.sh --help
```

```
Usage: /home/wangpf/bin/extractSeq.sh [--genome genome_file] [--gff3 gff3_file] [-
-cds cds_file] [--pep pep_file] [--id gene/mRNA_id] [--up up] [--down down] [--gz]
Options:
--genome    Specify the genome fasta file
--gff3      Specify the gff3 file
--cds       Specify the cds fasta file
--pep       Specify the pep fasta file
--id        Specify the gene/mRNA id
--up        Specify how many bp upstream for gene/mRNA
--down      Specify how many bp downstream for gene/mRNA
--gz        Compress all the result if this option is present
--help      Display this help message
```

run_edgeR.R

A R script for differential expression analysis using edgeR (without biological replication). You need to prepare three files:

Requiemment

- **read count matrix file.**
- **samples file**, tab-delimited text file indicating biological replicate relationships. e.g.

```
cond_A  sample_A
cond_B  sample_B
```

- **contrasts file**, tab-delimited text file containing the pairs of sample comparisons to perform. e.g.

```
cond_A  cond_B
cond_Y  cond_Z
```

Usage

```
Rscript run_edgeR.R -h
```

```
usage: run_edgeR.R [--] [--help] [--opts OPTS] [--matrix MATRIX]
      [--samples_file SAMPLES_FILE] [--min_reps MIN_REPS] [--min_cpm
      MIN_CPM] [--contrasts CONTRASTS] [--dispersion DISPERSION]
```

Run differential expression analysis using DESeq2.

flags:

-h, --help show this help message and exit

optional arguments:

-x, --opts RDS file containing argument values
-m, --matrix matrix of raw read counts (not normalized!)
-s, --samples_file tab-delimited text file indicating biological
 replicate relationships.
--min_reps At least min count of replicates must have cpm
 values > min cpm value. [default: 1]
--min_cpm At least min count of replicates must have cpm
 values > min cpm value. [default: 1]
-c, --contrasts file (tab-delimited) containing the pairs of
 sample comparisons to perform.
-d, --dispersion edgeR dispersion value. [default: 0.1]

get_longest_seq.pl

In some genomics data, there are multiple isoforms for one gene because of alternative splicing. This perl script can get the longest CDS or pep sequence of genes.

Requiemment

- perl module
 - Bio::SeqIO
- data file
 - CDS or pep sequence file
 - gff file

Usage

```
perl get_longest_seq.pl -h
```

```
#####
#
# Usage:  /public/home/wangpf/bin/get_longest_seq.pl --fasta <cds_or_pep.fa> --gff
#         <genes.gff> --out <outprefix>
#
# Required:
#
#         --fasta <string>                CDS or pep fasta file.
#
#         --gff    <string>                gff file.
#
#         --out    <string>                output prefix.
#
#####
```

Output

- <outprefix>.longest.fa, the longest CDS or pep sequence in fasta format with gene id as sequence identifier
- <outprefix>.longest.list, a list contain gene id, longest mRNA id and the length of longest mRNA

find_gaps.py

There are gaps in majority genome fasta files except T2T genome. This `find_gaps.py` script will find gaps position.

Requiemment

- python3 module
 - argparse
 - Biopython
- input data
 - genome fasta file

Usage

```
chmod u+x find_gaps.py
find_gaps.py -h
```

```
usage: find_gaps.py [-h] -i INPUT -o OUTPUT [-s] [-c CONTIG]
```

Find gaps (N regions) in a genome FASTA file and optionally split at gaps.

optional arguments:

```
-h, --help            show this help message and exit
-i INPUT, --input INPUT
                        Input genome FASTA file
-o OUTPUT, --output OUTPUT
                        Output file for gap positions
-s, --split           Split the FASTA file at gaps
-c CONTIG, --contig CONTIG
                        Output file for split contigs (default: split.fa)
```

Output

A tab-separated text file containing three columns.

```
<chromosome_id> <gap_start> <gap_end>
```

A fasta file with split contigs.

mummerCoordsPlot.R

`mummerCoordsPlot.R` has been integrated into `AlignmentDotPlot.R`.

This script can draw an alignment plot according to mummer's show-coords program. In fact, this script is modeled after `mummerCoordsDotPlotly.R` in `dotPlotly`. `mummerCoordsDotPlotly.R` is an excellent program, but there are a few aspects that I am not entirely satisfied with:

- It will sort query ID by length, but I want to keep query ID order in the query genome;
- `Yellow` is not very noticeable on a white background, so I use `red` and `blue`;
- Add the ability to color the alignment results based on whether they are forward or reverse complementary;
- IDs were not on the middle of chromosomes or contigs in figure;
- I typically work with the `tidyverse` ecosystem, so I have rewritten this script using `tidyverse` syntax for clarity and consistency;
- Some other modifications.

AlignmentDotPlot.R

`AlignmentDotPlot.R` is an improved and extended version of `mummerCoordsPlot.R`.

It retains all original functionalities for visualizing **MUMmer** alignment results, and adds new support for **minimap2** outputs in **PAF** format.

This script enables users to generate high-quality alignment dot plots from both **MUMmer** and **minimap2**, providing a unified and convenient workflow for comparing genome assemblies or large-scale sequence alignments.

Requiemment

- R package
 - argparser
 - tidyverse
- input data
 - Alignment result from mummer's show-coords program:

```
nucmer --mum --prefix=example -g 1000 -c 90 -l 40 -t 10
GWHERGL000000000.genome.fasta GWHERGM000000000.genome.fasta > nucmer.log
delta-filter -r -q example.delta > example.filter
show-coords example.filter.delta > example.coords
```

- or, Alignment result from minimap2:

```
minimap2 -x asm5 -o example.paf -t 20 GWHERGL000000000.genome.fasta
GWHERGM000000000.genome.fasta
```

- Tab-separated text files containing reference and query IDs separately, first column is IDs in genome, the second is what you want to show in figure, only chromosomes or contigs in these files will be shown (optional). If not assigned, all IDs of chromomsomes or contigs longer than **--min-query-length** will be shown in figure, or you can just assign in the command line, e.g. "GWHERGL000000001:A01,GWHERGL000000002:A02,GWHERGL000000003:A03"

```
GWHERGL000000001 A01
GWHERGL000000002 A02
GWHERGL000000003 A03
GWHERGL000000004 A04
GWHERGL000000005 A05
... ..
```

- Tab-separated text files containing chromosome or contig length, first column is IDs in genome, the second is the corresponding length (optional);

```
GWHERGL000000001 59205763
GWHERGL000000002 39005548
GWHERGL000000003 44769757
GWHERGL000000004 28346059
GWHERGL000000005 46671336
... ..
```

Usage

```
Rscript AlignmentDotPlot.R -h
```

```
usage: AlignmentDotPlot.R [--] [--help] [--opts OPTS] [--tool TOOL]
      [--input INPUT] [--out OUT] [--ref REF] [--query QUERY]
      [--refIDs REFIDS] [--queryIDs QUERYIDS] [--refLen REFLen]
      [--queryLen QUERYLen] [--min-query-length MIN-QUERY-LENGTH]
      [--min-alignment-length MIN-ALIGNMENT-LENGTH] [--min-identity
      MIN-IDENTITY] [--min-mapQ MIN-MAPQ] [--color-by COLOR-BY]
      [--size SIZE] [--width WIDTH] [--height HEIGHT]
```

Generates plots of alignment data produced by show-coords.

flags:

-h, --help show this help message and exit

optional arguments:

-x, --opts	RDS file containing argument values
-t, --tool	alignment tool, 'mummer' or 'minimap2'
-i, --input	coords file from mummer program
	'show-coords' or paf file from minimap2
-o, --out	outfile prefix
-r, --ref	reference name shown in plot [default: Reference]
-q, --query	query name shown in plot [default: Query]
-R, --refIDs	a tab-separated file containing two columns, first column is IDs in reference genome, second column is names to shown in plot, or a string like 'scaffoldA01:A01,scaffoldA03:A03'. If there is no file named that, we will threat it as a string
-Q, --queryIDs	a tab-separated file containing two columns, first column is IDs in query genome, second column is names to shown in plot, or a string like 'scaffoldA01:A01,scaffoldA03:A03'. If there is no file named that, we will threat it as a string
-l, --refLen	a tab-separated file containing two columns, first column is IDs in reference genome, second column is chr length
-L, --queryLen	a tab-separated file containing two columns, first column is IDs in query genome, second column is chr length
-M, --min-query-length	filter queries with total alignments less than cutoff X bp [default: 4e+05]
-m, --min-alignment-length	filter alignments less than cutoff X bp

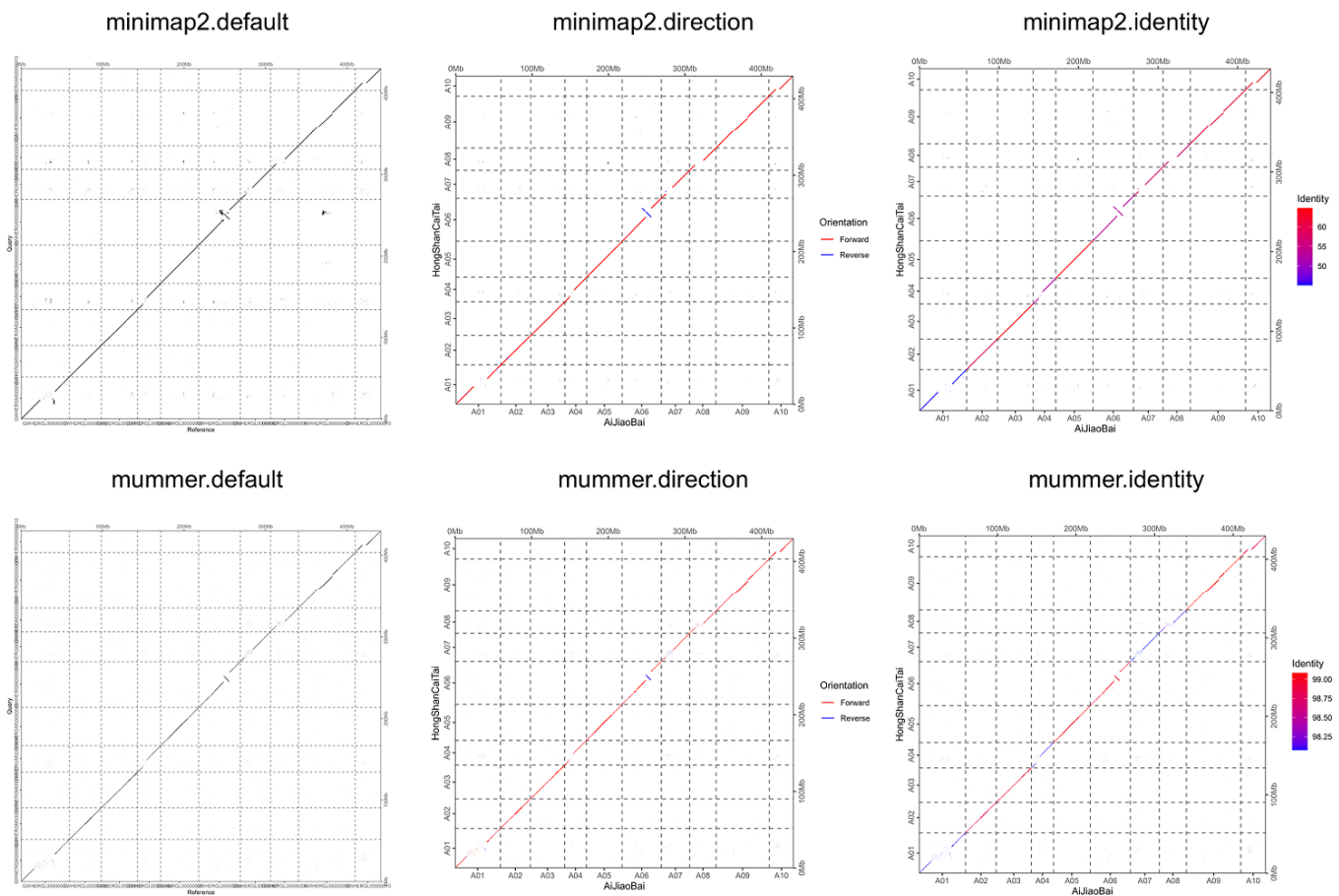
	[default: 2000]
-s, --min-identity	filter alignments with identity less than X % [default: 0]
-p, --min-mapQ	minimum mapQ, is effective only for minimap2 [default: 0]
-c, --color-by	turn on color alignments by 'direction' or 'identity', no color if not assign
-S, --size	line width of alignments in figure
-W, --width	plot width (inches) [default: 10]
-H, --height	plot height (inches) [default: 10]

Output

```
# mummer
## default
Rscript AlignmentDotPlot.R -t mummer \
  -i example.coords -o example.default
## direction
Rscript AlignmentDotPlot.R -t mummer \
  -i example.coords -o example.direction \
  -r AiJiaoBai -q HongShanCaiTai \
  -R refIDs.txt -Q queryIDs.txt \
  -l refLen.txt -L queryLen.txt \
  -m 5000 -s 90 \
  -c direction \
  -W 7 -H 6
## identity
Rscript AlignmentDotPlot.R -t mummer \
  -i example.coords -o example.identity \
  -r AiJiaoBai -q HongShanCaiTai \
  -R refIDs.txt -Q queryIDs.txt \
  -l refLen.txt -L queryLen.txt \
  -m 5000 -s 90 \
  -c identity \
  -W 7 -H 6

# minimap2
## default
Rscript AlignmentDotPlot.R -t minimap2 \
  -i example.paf -o example.default
## direction
Rscript AlignmentDotPlot.R -t minimap2 \
  -i example.paf -o example.direction \
  -r AiJiaoBai -q HongShanCaiTai \
  -R refIDs.txt -Q queryIDs.txt \
  -l refLen.txt -L queryLen.txt \
  -m 5000 -p 60 \
  -c direction \
  -W 7 -H 6
## identity
```

```
Rscript AlignmentDotPlot.R -t minimap2 \
-i example.paf -o example.identity \
-r AiJiaoBai -q HongShanCaiTai \
-R refIDs.txt -Q queryIDs.txt \
-l refLen.txt -L queryLen.txt \
-m 5000 -p 60 \
-c identity \
-W 7 -H 6
```



Data Source

These two genome is from this paper:

Zhou, Yifan et al. "The complexity of structural variations in Brassica rapa revealed by assembly of two complete T2T genomes." Science bulletin vol. 69,15 (2024): 2346-2351. doi:10.1016/j.scib.2024.03.030

BackgroundAnalysis.R

It is important to know the length and position of introgression loci in marker-assisted selection, and this R script is used for genetic background recovery rates analysis.

Requiemment

- [SNPbinner](#) (Python 3)
- R package

- argparser
- tidyverse
- cowplot
- ggh4x
- ggtrastr (optional)
- Tab-separated text file containing chromosome ID and labels shown in the result. (optional)
- Tab-separated text file containing chromosome ID and length. (optional)

Usage

```
Rscript BackgroundAnalysis.R -h
```

```
usage: BackgroundAnalysis.R [--] [--help] [--opts OPTS] [--input INPUT]
      [--donor DONOR] [--recurrent RECURRENT] [--sample SAMPLE]
      [--chromosome CHROMOSOME] [--length LENGTH] [--minDdp MINDDP]
      [--maxDdp MAXDDP] [--minRdp MINRDP] [--maxRdp MAXRDP]
      [--path_to_snpbinner PATH_TO_SNPBINNER]
```

a program for background analysis

flags:

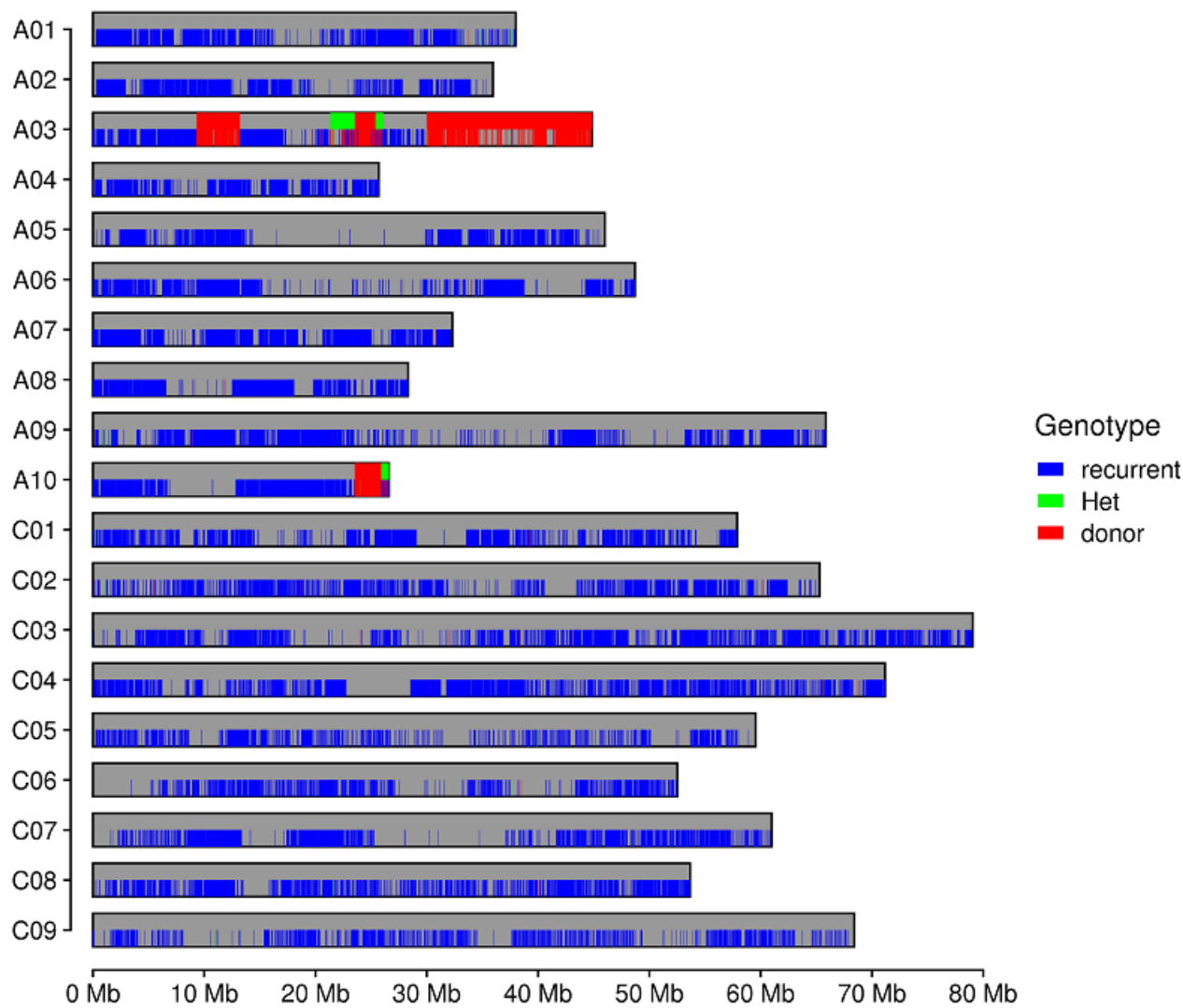
-h, --help show this help message and exit

optional arguments:

-x, --opts	RDS file containing argument values
-i, --input	input file name, GATK Table format
-d, --donor	donor parent name
-r, --recurrent	recurrent parent name
-s, --sample	sample parent name
-c, --chromosome	tab-separated file contain chromosome ID and label shown in figure. if not assigned, all chromosome in genome will be shown
-l, --length	tab-separated file contain chromosome ID and length. if not assigned, the max position of SNP for each chromosome in GATK Table file will be regarded as chromosome length
-m, --minDdp	Minimum depth for donor parent [default: -Inf]
--maxDdp	Maximum depth for donor parent [default: Inf]
--minRdp	Minimum depth for recurrent parent [default: -Inf]
--maxRdp	Maximum depth for recurrent parent [default: Inf]
-p, --path_to_snpbinner	path to snpbinner [default: ~/tools/SNPbinner-1.0.0-GondaEtAl2019/snpbinner]

Output

- a figure which show the length and position of introgression loci.



- genetic background recovery rate, <SAMPLE>.bg_stat.csv.

genotype	length(bp)	rate
Donor	23013590	0.023952016
Heterozygous	3605578	0.003752603
Recurrent	934201436	0.972295382

count_gff_length.py

Calculate the total length of a specific feature type in a GFF file, e.g. exon, gene, rRNA.

Usage

```
python count_gff_length.py -h
```

```
usage: count_gff_length.py [-h] -g GFF -c CATEGORY
```

Calculate total length and count of a specific feature in a GFF file.

optional arguments:

```
-h, --help            show this help message and exit
-g GFF, --gff GFF     Input GFF file
-c CATEGORY, --category CATEGORY
                        Feature type to calculate length and count (e.g.,
                        gene, mRNA, miRNA)
```

Input

A gff file.

find_telomere_from_TRF.py

This script is used for identifying telomere from [TRF \(Tandem Repeats Finder\)](#). **Please note that the TRF output must first be converted into GFF format using [TRF2GFF](#).**

Usage

```
python find_telomere_from_TRF.py -h
```

```
usage: find_telomere.py [-h] -i INPUT -o OUTPUT [-m MOTIF] [-c COPY] [--match
MATCH] [--indel INDEL]
```

Filter telomere tandem repeats from TRF gff3

optional arguments:

```
-h, --help            show this help message and exit
-i INPUT, --input INPUT
                        Input GFF3
-o OUTPUT, --output OUTPUT
                        Output file
-m MOTIF, --motif MOTIF
                        Telomere motif (default: TTTAGGG)
-c COPY, --copy COPY  Minimum CopyNumber (default: 50)
--match MATCH         Minimum PercentMatches (default: 0)
--indel INDEL         Maximum PercentIndels (default: inf)
```

Output

A gff format file containing telomere.

```
trf genome.fa 2 7 7 80 10 50 500 -f -d -m  
trf2gff -i genome.fa.2.7.7.80.10.50.500.dat -o genome.TRF.gff3  
find_telomere_from_TRF.py -i genome.TRF.gff3 -o telomere.gff3 --match 90
```